

Qatar-4b

R-2706

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_180616 · created 2026-09-01T00:45:18+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458285.90009 +/- 0.00031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.053
Transit depth (Rp/Rs)^2	3.14 +/- 1.87 [%]
Orbital Inclination (inc)	88.6 +/- 2.13
Airmass coefficient 1 (a1)	1.2 +/- 0.72
Airmass coefficient 2 (a2)	-0.18 +/- 0.31
Scatter in the residuals of the lightcurve fit is	57.12 %
Best Comparison Star	#2 - [418, 142]
Optimal Aperture	1.46
Optimal Annulus	4.25
Transit Duration (day)	0.0921 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.053 vs 0.138 (deviation 0.72σ)
Duration fit vs published	0.0921 d vs 0.0892 d (deviation 0.42σ)
Mid-time O–C	-1.0 min
Depth SNR	3.3
Residual scatter	57.12 %

Light curve

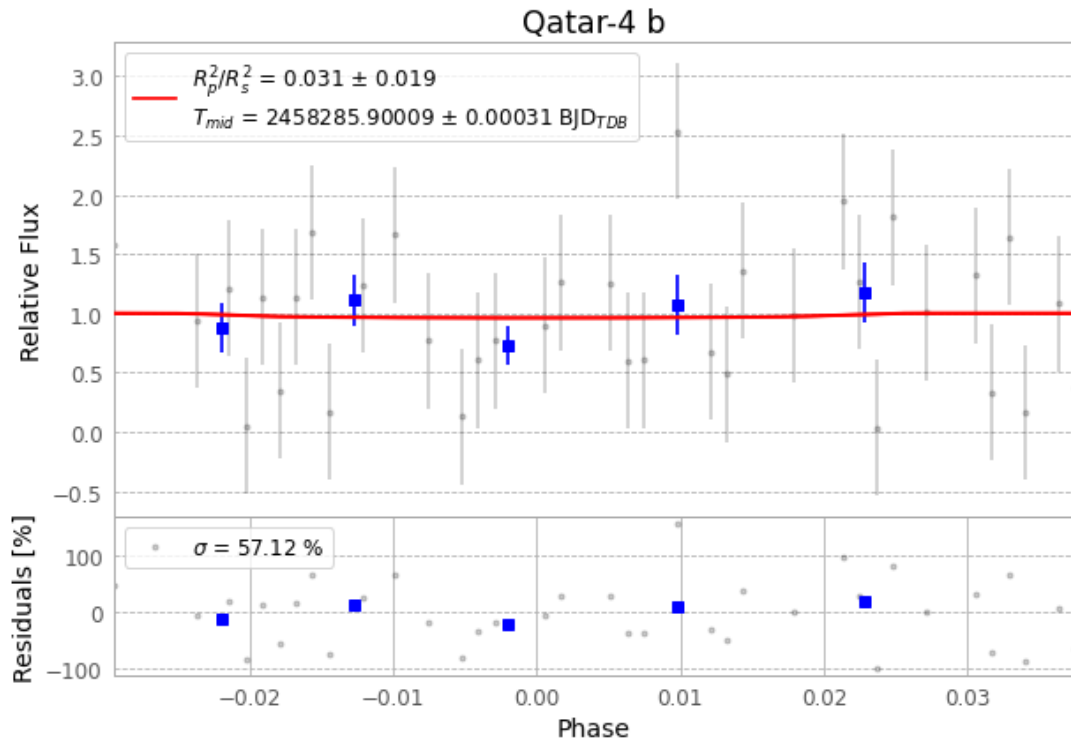


Figure 1 — FinalLightCurve_Qatar-4 b_2018-06-16.png (SHA-256 9bbf89887651b180...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 139], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8c6693c204aa545e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-4180616081514.FITS → Qatar-4180616111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-180616.log (79d46d973412...), FinalLightCurve_Qatar-4 b_2018-06-16.png (9bbf89887651...), FinalLightCurve_Qatar-4 b_2018-06-16.csv (c049286d12c6...)

Compute resources

Wall time 112.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:45Z.